

BIOL-1 Exam 03

College of the Redwoods, Del Norte

Instructor: Dr. Daniel Friedman

Name: _____

Date: _____

Modules 12-15

Total Points: 50

Part A: Multiple Choice (30 points)

Select the best answer for each question. Each question is worth 1 point.

Module 12: Darwin and Evolution

1. Darwin's phrase **descent with modification** best captures the idea that:
 - A) Each species was created in its present form and cannot change
 - B) Only the strongest individual in a species updates the gene pool during its lifetime
 - C) Related lineages share ancestors and accumulate heritable change over many generations
 - D) Evolution requires organisms to choose beneficial mutations on purpose

2. In evolutionary biology, **fitness** is most appropriately defined as:

- A) Relative contribution of genes to the next generation compared with others in the population
- B) Maximum possible lifespan regardless of reproduction
- C) Ability to learn new behaviors within one lifetime
- D) Total body mass relative to other species

3. Which observation in Darwin's reasoning corresponds to **overproduction of offspring**?

- A) Parents vary in traits that can be inherited
- B) Many descendants are produced, but not all survive and reproduce
- C) Populations never change allele frequencies
- D) Mutations always match what the environment "needs"

4. **Homologous** features such as the forelimb bones of a whale, a bat, and a human support evolution because they:

- A) Show identical functions in every species
- B) Reflect shared ancestry with structural modification for different roles
- C) Prove that whales evolved from modern bats
- D) Demonstrate that analogous traits never evolve

5. Convergent evolution produces superficially similar forms in distant lineages mainly because:

- A) Gene flow merges unrelated species
- B) Strong directional selection removes all variation instantly
- C) Similar selection pressures favor similar phenotypes even without recent shared ancestry
- D) Hardy–Weinberg equilibrium forces identical traits

6. Which statement best explains why biologists say **populations** evolve rather than **individuals**?

- A) Individuals cannot change any traits during life

- B) Evolution is measured as change in allele frequencies in a population across generations
 - C) Natural selection ignores heritable variation
 - D) Fitness applies only to entire ecosystems, not organisms
7. A tan deepens on human skin after UV exposure; the change is not reliably inherited at the same intensity by children who were never exposed. This scenario highlights:
- A) Evolutionary fitness increasing in one generation
 - B) Acclimation within a lifetime versus allele-frequency change across generations
 - C) Founder effect in humans
 - D) Sympatric speciation
8. Related finch species on neighboring islands with differing beaks illustrate which broad line of evidence when paired with geographic patterns?
- A) Comparative biochemistry only rules out fossils
 - B) Biogeography combined with descent with modification
 - C) Proof that drift never occurs on islands
 - D) Evidence that species definitions never use reproduction

Module 13: How Populations Evolve

1. **Microevolution**, as used in this course, refers to:
- A) Growth of one organism from juvenile to adult size
 - B) Change in allele frequencies in a population over time
 - C) Sudden origin of a new kingdom in one mating event
 - D) Replacement of DNA by RNA in every cell
2. Alleles enter or leave a population through migration and interbreeding with other populations in the process called:
- A) Genetic drift
 - B) Founder effect only

- C) Gene flow
 - D) Postzygotic isolation
3. A hurricane randomly kills most individuals in an insect population; survivors rebuild the population with altered allele frequencies largely due to chance sampling. This scenario fits:
- A) Directional selection on camouflage every time
 - B) Bottleneck effect associated with genetic drift
 - C) Increased gene flow through the storm
 - D) Hardy–Weinberg equilibrium maintenance
4. A handful of rodents reach a new habitat and found a population whose gene pool reflects only the alleles those founders carried. This illustrates:
- A) Bottleneck in the source population only
 - B) Founder effect (genetic drift)
 - C) Disruptive selection on teeth without variation
 - D) Sympatric polyploid speciation
5. When **intermediate** human birth weights survive best and **extremes** are selected against, the pattern is labeled:
- A) Directional selection
 - B) Disruptive selection
 - C) Stabilizing selection
 - D) Genetic drift only
6. When **both extremes** of a trait survive and reproduce better than **intermediates**, the pattern is:
- A) Stabilizing selection
 - B) Directional selection only
 - C) Disruptive selection
 - D) Founder effect

7. The Hardy–Weinberg equilibrium model is useful primarily because it provides:
- A) Proof that evolution never occurs in nature
 - B) A mathematical baseline against which real populations can be compared
 - C) The formal definition of macroevolution only
 - D) Carrying capacity estimates for every species
8. Mutations originate new alleles; together with gene flow, genetic drift, non-random mating, and natural selection, they describe processes that can:
- A) Guarantee allele frequencies never move from one generation to the next
 - B) Change allele frequencies in populations
 - C) Eliminate all variation immediately in large populations
 - D) Replace sunlight as ecosystem energy input

Module 14: Macroevolution

1. Under the **biological species concept**, species are recognized largely by:
- A) Matching color patterns in a field guide only
 - B) Identical chromosome numbers without breeding tests
 - C) Natural interbreeding within the group producing viable, fertile offspring and reproductive isolation from other such groups
 - D) Living in the same forest regardless of gene exchange
2. Two frog species breed in different months so hybrids rarely form. This barrier is:
- A) Postzygotic hybrid breakdown
 - B) Prezygotic temporal isolation
 - C) Density-dependent regulation
 - D) Exponential growth limitation
3. Weak or inviable hybrid offspring illustrate:
- A) Prezygotic behavioral isolation
 - B) Postzygotic reduced hybrid viability
 - C) Gene flow between populations

- D) Logistic carrying capacity

4. A sterile mule from horse × donkey matings represents:

- A) Prezygotic gametic incompatibility only
- B) Sympatric polyploid speciation
- C) Postzygotic reduced hybrid fertility
- D) Founder effect in horses

5. A new geographic barrier divides one interbreeding fish species into two populations that diverge until they no longer mate. This exemplifies:

- A) Sympatric speciation via polyploidy only
- B) Allopatric speciation
- C) Instant macroevolution without allele-frequency change
- D) Density-independent mutualism

6. A plant lineage doubles its chromosome number and becomes reproductively isolated **without** geographic separation. This route commonly illustrates:

- A) Allopatric divergence only
- B) Sympatric speciation (for example via polyploidy)
- C) Purely postzygotic prezygotic barriers
- D) Trophic cascade elimination

7. In this course, **macroevolution** is best understood as:

- A) A separate genetic mechanism unrelated to allele-frequency change
- B) Larger-scale evolutionary outcomes built from the same population-level processes accumulated over long spans
- C) Changes restricted to bacteria only
- D) Synonymous with acclimation within individuals

Module 15: Population and Systems Ecology

1. Unlimited-resource phases of population increase often follow a **J**-shaped curve described as:
 - A) Logistic growth approaching **K** immediately
 - B) Stable equilibrium without births or deaths
 - C) Hardy–Weinberg equilibrium
 - D) Exponential growth
2. In the logistic model, the symbol **K** stands for:
 - A) Carrying capacity
 - B) Mutation rate per generation
 - C) Number of trophic levels only
 - D) Hybrid sterility frequency
3. A contagious parasite spreads faster when hosts are crowded. That limiting factor is typically classified as:
 - A) Density-independent because parasites are small
 - B) Only abiotic regardless of biology
 - C) Density-dependent
 - D) Part of photosynthesis efficiency
4. A lightning fire kills a similar fraction of saplings in sparse and dense plots (idealized case where crowding does not change fire impact). Classified as a limiting disturbance in many texts, this behaves most like:
 - A) Density-dependent competition for mates only
 - B) An increase in **K** for every species
 - C) Conversion of heat back into chemical bond energy for producers
 - D) Density-independent mortality

5. The approximate **10% energy transfer** rule implies that supporting humans mainly on long marine food chains (many trophic steps) generally requires:

- A) More primary production at the base because energy is lost between levels
- B) Less primary production than eating plants directly
- C) Equal joules stored at every trophic level by definition
- D) Decomposers to create new sunlight inside consumers

6. In ecosystems, chemical nutrients such as nitrogen are recycled through producers and consumers, whereas energy captured by photosynthesis is ultimately:

- A) Stored unchanged inside fossils forever with no loss
- B) Returned photon-for-photon to chlorophyll in the same pathway as nutrient atoms
- C) Lost largely as heat along transfers through trophic levels
- D) Divided equally among all trophic levels without loss

7. Fungi and bacteria that reduce dead biomass and release usable nutrients back toward producers function ecologically as:

- A) Primary producers that fix carbon using sunlight only in soil
- B) Apex predators in every food chain
- C) Agents that recycle energy from heat into new chemical bonds for autotrophs
- D) Decomposers central to nutrient cycling

Part B: Fill in the Blank (11 points)

Questions 1–11. Complete each statement with the best biological term from the word bank. Each question is worth 1 point.

Word bank: Allele frequencies, Allopatric, Decomposers, Directional selection, Gene flow, Genetic drift, Hardy–Weinberg, Homologous, Postzygotic, Prezygotic, Sympatric

1. Formal **microevolution** focuses on change in _____ from one generation to the next within a population.

2. Migration and mating can move alleles between populations; together this process is called _____.
3. Random shifts of allele frequencies—especially pronounced when population size is small—reflect _____.
4. Forelimb bones built on the same anatomical plan but used for flight, swimming, or grasping are classic examples of _____ structures from shared ancestry.
5. A geographic barrier separates populations that diverge without exchanging genes; speciation starting this way is often _____ speciation.
6. Instant reproductive isolation via chromosome doubling in plants in the same region exemplifies _____ speciation in many textbook examples.
7. Isolating mechanisms such as different mating seasons that act **before** a zygote forms are _____ barriers.
8. Reduced fertility of hybrids such as mules classifies as a _____ barrier to gene flow.
9. The equilibrium equation used as a **null** baseline assuming no evolution is associated with the names _____.
10. When soot favors darker moths and the population average shifts toward one extreme of color variation, the pattern is often _____.
11. Organisms that recycle elements from detritus back to soil nutrients available to producers include soil fungi and many bacteria called _____.
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Part C: Free Response (9 points)

Choose THREE of the following five questions. Answer clearly in the space provided. Each response is worth 3 points.

1. **Natural Selection and Evidence.** State what biologists mean by **natural selection** using the ideas of heritable **variation**, **overproduction** of offspring, and **differential**

- reproductive success.** Then name **one** type of evidence for evolution (fossil, comparative anatomy, biogeography, or molecular) and state briefly what pattern it highlights.
2. **Drift, Flow, and Selection.** Contrast **genetic drift** with **natural selection** (what drives allele-frequency change in each?). Give **one** plain-language example of a **bottleneck** or **founder** effect. Finally, explain **gene flow** in one sentence.
3. **Reproductive Isolation and Speciation.** Define **reproductive isolation** in your own words. Give **one** example of a **prezygotic** barrier and **one** example of a **postzygotic** barrier. Contrast **allopatric** versus **sympatric** speciation in **two** sentences or fewer.
4. **Population Growth and Regulation.** Compare **exponential** versus **logistic** population growth curves and state what **K** represents. Then give **one** example of a **density-dependent** limiting factor and **one** example of a **density-independent** disturbance (each in a full sentence).
5. **Energy and Nutrients.** Contrast **energy flow** through trophic levels with **nutrient cycling** (include producers and decomposers). Use the **~10% rule** to explain why diets rich in **high trophic-level** fish can demand **more** primary production than feeding humans closer to **primary producers**.

Response 1

Question Number: _____

Response 2

Question Number: _____

Response 3

Question Number: _____

Feedback (Optional)

A. Is there anything else you would like to share regarding your exam preparation, or concepts you found particularly interesting/challenging?

B. What score (or percentage) do you estimate you earned on this exam? How confident are you in that estimate (1–10)?

End of Exam 03